Josephus Problem I

Time limit: 1.00 s
Memory limit: 512 MB

CSES 2162 | Sorting and Searching | https://cses.fi/problemset/task/2162

Consider a game where there are n children (numbered 1,2,…,n) in a circle. During the game, every other child is removed from the circle until there are no children left. In which order will the children be removed?

Input

The only input line has an integer n.

Output

Print n integers: the removal order.

Constraints

- $1 \le n \le 2 \cdot 10^5$

Example

Input
7

Output
2 4 6 1 5 3 7